

MAPPING THE SKY

TO FIND OBJECTS IN SPACE AND TO MAKE MAPS OF THE SKY, ASTRONOMERS USE A FRAME OF REFERENCE CALLED THE CELESTIAL SPHERE. THIS SPHERE IS AN IMAGINARY SHELL, CENTERED ON THE EARTH, UPON WHICH ANY OBJECT IN THE SKY CAN BE LOCATED.

We know that objects in space can lie at any distance from Earth, but in order to position them on a map we can think of them as all being stuck to the inside of the celestial sphere. Just like the Earth itself, the sphere can be divided up with lines of longitude and latitude, including an equator. Similarly, just as the land area of the Earth is separated into countries, the celestial sphere is divided into areas called constellations.

► The constellations

For millennia, humans have joined stars with imaginary lines to make recognizable patterns, or constellations. These patterns include the outlines of animals and mythical beasts and heroes. In the early 20th century, the International Astronomical Union gave formal definition to 88 constellations, giving them official names and setting the positions of their boundaries. In this modern system, a constellation is an area of sky rather than a pattern of lines between stars.

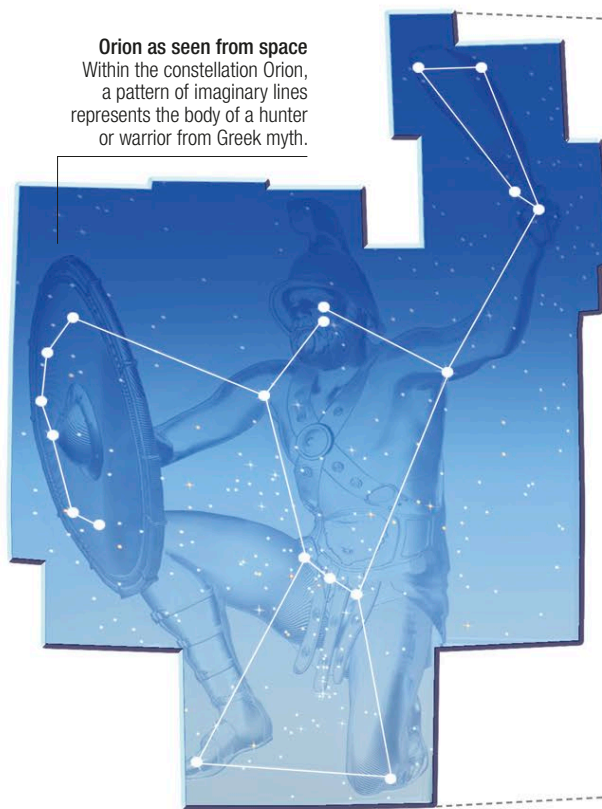
▼ Observer's location

From a particular place on Earth, up to half of the celestial sphere can be seen at any one time, with the rest hidden by the Earth itself. Whether or not a particular constellation is visible also depends on an observer's location. For example, all of the constellation Canis Major can be seen between latitudes 56 degrees north and the south pole. From a belt to the north of this, only part of the constellation can be seen, while in the region around the north pole, none of the constellation is visible.

The celestial sphere is an imaginary sphere surrounding Earth

Constellation boundaries are straight and either horizontal or vertical

Orion as seen from space
Within the constellation Orion, a pattern of imaginary lines represents the body of a hunter or warrior from Greek myth.



CANIS MAJOR VISIBILITY FROM EARTH

Canis Major

Constellation not visible

Part of constellation visible

Whole constellation visible